



Polypropylene nonwoven fabrics with conformational grafting of poly(glycidyl methacrylate) for bioseparations

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ABSTRACT

Nonwovens fabrics are porous materials with great potential for use in the separation and purification of biotherapeutic proteins and other biomolecules after proper surface activation and modification. This work describes an activation process for polypropylene nonwoven membranes by conformational coating of poly(glycidyl methacrylate) (polyGMA) through a UV pretreatment–UV grafting process (UV–UV) in the presence of benzophenone as initiator. The grafting mechanism relies on the enhanced adsorption of benzophenone to the fiber surface after the UV pretreatment. It was found that this process results in highly conformational and uniform polyGMA grafts on the surface of the PP nonwoven fibers. After grafting, a hydrophilic spacer, diethylene glycol, an anion exchange ligand, diethyl amine, and a combination of diethylene glycol spacer–primary amine were attached to the PP nonwoven surface with the conformational polyGMA grafts. High equilibrium protein binding capacities for bovine serum albumin (BSA) under static and flow conditions were obtained (120 mg/g and 102 mg/g, respectively). A column packed with modified nonwovens exhibited higher permeability coefficient ($1.08 \times 10^{-7} \text{ cm}^2$) than columns packed with chromatographic resins. The total porosity (to acetone) was found to be 0.80 and the interstitial porosity (to BSA) was 0.48, measured by pulse experiments using acetone and BSA respectively as tracers under nonbinding conditions.

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1. Introduction

Industrial biotherapeutic protein isolation and purification processes are conventionally carried out using packed-bed chromatography columns packed with porous resins that have ion exchange or affinity ligand functionalities. Chromatographic resins tend to exhibit diffusional limitations inside the particle pores, relatively large pressure drops and high cost [1,2]. The diffusional limitations tend to limit the dynamic binding capacity of the resins to capture product, and the column format reduces throughput as a result of the increased pressure drop at higher flow rates. As a result, the biopharmaceutical industry is seeking alternative materials to replace traditional packed-bed column resins for the purification of biotherapeutics [3–16]. One alternative is to use ion exchange and affinity membranes made of nonwoven fabrics which exhibit variable and controllable porosity, fiber diameter and pore sizes [17]. These nonwoven fabrics are made of very inexpensive materials such as polypropylene, poly(butylene terephthalate) and nylon. With nonwoven fabrics, ion exchange or affinity ligands could be directly attached to the fiber surface, thus reducing diffusional lim-

itations as the process fluid is transported across the membrane. In addition, the pore size and overall porosity of the membrane can be controlled to operate at very low pressure drops, resulting in increased throughput compared to column operations. The production costs of nonwovens fabrics are very low and the materials are relatively inexpensive. As a result, nonwovens also offer the opportunity to develop disposable bioseparation devices.

Despite the potential advantages of using nonwoven fabrics as separation media, there are several barriers to overcome before they can be used in commercial operations. The surface of most available nonwoven fibers is generally hydrophobic and not functionalized, leading to a significant amount of nonspecific binding of proteins and contaminants that can limit product purity. Surface activation is indispensable, not only to reduce nonspecific proteins adsorption, but also to allow for the covalent attachment of ion exchange or ligand functionalities to adsorb the desired product or contaminant. In addition, for product capture, it is necessary to enhance the surface area available for protein product binding by reducing the fiber size to sub-micron diameters.

This paper describes a method of activation of the surface of PP nonwoven fabrics that produces highly conformational poly(glycidyl methacrylate) (polyGMA) grafts using a UV pretreatment step and a UV grafting step with benzophenone as an initiator. The epoxy end groups in polyGMA enable the covalent attachment of a large num-

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ber of chemical ligands, including hydroxyls, amines, carboxylic acids, pyridines, and peptides [7]. In this process the polymer grafts conform to the curvature of the fibers within the fabric mat, achieving full coverage with a consistent thickness of the grafted polymer layer throughout the sample. Using this method, it is possible to avoid pore blockage that can change the transport and adsorption properties of the nonwoven after activation. In the literature, only a few studies can be found on the activation of nonwoven membrane fabrics by surface grafting for bioseparations [18–21]. In these earlier works, the issue of conformity of the resulting grafts has not been addressed, so this is the first paper to deal with this issue directly.

Following the grafting with poly(GMA), the activated PP nonwoven was further modified with ligands and spacers with potential applications in bioseparations. In the first example, the nonwoven was modified to be a strong anion exchange membrane by the coupling of diethyl amine (DEA). In the second example, the nonwoven was modified to reduce nonspecific binding of proteins by the coupling of a diethylene glycol (DEG) spacer. In the third example, a primary amine was attached to the DEG spacer by reaction with epichlorohydrin and ammonium hydroxide. The primary amine is primarily used for the coupling of affinity ligands. It also has the capability of being a weak anion exchanger. Equilibrium protein binding capacities for each sample were measured using bovine serum albumin (BSA) under static conditions. Binding capacities were also measured under flow conditions for the nonwoven fabrics modified with tertiary and primary amines. In addition, the total and interstitial porosities and permeability of the nonwoven modified with DEG were measured. The results show that the equilibrium binding capacities of the nonwoven membranes can be as high as 120 mg/g and 102 mg/g for static and flow conditions. Because of the high porosity of these membranes, pressure drop measurements indicate that these high binding capacities are achieved with higher flow permeabilities than columns of ion exchange resin with similar or lower capacities. The results of this work clearly indicate that nonwoven fabrics membranes have high binding capacities and low pressure drops that make them attractive alternatives to other types of membranes and particles currently being used in industrial bioseparations.

2. Experimental

2.1. Materials

The polypropylene nonwoven fabric (PP 175, Macopharma, Tourcoing, France) used in this study was produced from melt-blowing process. Its density and porosity are 0.18 g/cm³ and 0.80, respectively. The porosity of the nonwoven was calculated from the differences between the densities of the PP nonwoven fabrics and pure polypropylene. The average pore size of the nonwoven is 11 ± 4 μm. It was measured by a Capillary Flow Porometer (Model CFP-1100-AX, Porous Materials Inc., Ithaca, NY). Fiber diameters of the nonwovens were in the range of 1–20 μm with an average diameter of around 8 ± 4 μm. This value was obtained by measuring fiber diameters from SEM images. The average thickness of the nonwoven is 0.2 mm. The crystallinity of the fiber is 52% as determined by Differential Scanning Calorimetry (Q-100 DSC, TA Instruments, New Castle, DE). The nonwoven fabric samples were washed in acetone, methanol and water, and dried in an oven prior to use. The dried samples were cut into 2 cm × 4 cm strips before the grafting procedure. The total surface area per mass of nonwoven fabric estimated from the average fiber diameter value is within 0.3–0.5 m²/g.

Glycidyl methacrylate (GMA) monomer, bovine serum albumin (BSA), diethylene glycol (DEG) and epichlorohydrin (EPI) and

diethylamine (DEA) were purchased from Sigma–Aldrich (St. Louis, MO). A pre-packed Sigma inhibitor remover column was used to remove any reaction inhibitors prior to use (Product: 306320, St. Louis, MO). Butanol and ammonium hydroxide were obtained from Fisher Chemical (Fairlawn, NJ; Grade: ACS). Tetrahydrofuran (THF) and methanol were obtained from Sigma–Aldrich (St. Louis, MO). All solvents were used as received without further purification. Benzophenone was ordered from Sigma–Aldrich (St. Louis, MO).

2.2. Determination of activation/grafting conditions

Polypropylene nonwoven was first pretreated by exposure to UV light of 185/254 nm wavelength in air. This was followed by UV-initialized grafting of GMA at 365 nm in the presence of benzophenone. UV pretreatment was conducted with a UV lamp (Model 42, Jelight, Irvine, CA) of intensity ~50 mW/cm² and the grafting step was carried out with a UV lamp (EA-180, Spectroline, Westbury, NY) of intensity ~5 mW/cm². Each pretreated sample was sprayed with 0.2 ml 20% (v/v) GMA in butanol solution containing 1.3% (w/w) benzophenone (initiator to monomer ratio = 1:25). The sample was then sandwiched between two glass slides and put under the UV lamp for grafting with the pretreated side of the sample exposed to the UV. After grafting, unreacted monomer, homopolymer and benzophenone were removed from the sample by sonication in THF and methanol, each for 30 min, sequentially. The sample was then dried in air and analyzed by FTIR-ATR and SEM. The kinetics of the grafting process was measured by conducting experiments for different pretreatment and grafting times at the above concentration. In addition, grafting was also conducted with a 20% monomer solution containing 6.5% (w/w) benzophenone (initiator to monomer ratio = 1:5) for a single pretreatment and grafting times.

The effect of UV pretreatment on the adsorption of benzophenone to the nonwoven fiber surface was measured by immersing nonwovens pretreated by UV for various times into a 1.3% (w/w) benzophenone in butanol solution. The immersion times were 1, 10, 15 and 30 min. UV irradiation was not employed during the immersions. After immersion, the samples were taken out, hard-pressed between two paper towels (Wypall X60, Kimberley Clark, Roswell, GA) to remove the solution trapped in the pores, dried in air and analyzed by Fourier Transform InfraRed Spectroscopy-Attenuated Total Reflection (FTIR-ATR).

The grafting solution consisting of benzophenone and GMA in butanol shows a strong absorbance peak at 240 nm (UVC) and a weak absorption peak at 340 nm (UVA), which were measured by UV–vis spectrometer (Nicolet evolution 300). Due to the use of glass slides as holders for grafting in this experiment, UVA was selected for its higher transmission through glass than UVC.

2.3. Coupling of ligands to the activated nonwoven

The nonwovens grafted for ligand attachment were prepared by pretreatment for 10 min and grafting for 15 min on both sides of the nonwoven. This leads to less spatial variation of grafting inside the nonwoven. Tertiary amine was attached to the activated nonwoven surface by the immersion of the nonwoven into 50% diethylamine water solution at 30 °C for 5 h. Tertiary amine was formed by the ring-opening reaction between diethylene amine and epoxy. Diethylene glycol (DEG) spacer was attached to the activated nonwoven via the ring-opening reaction of epoxy. The nonwoven was immersed in pure DEG with addition of catalytic amounts of sulfuric acid [22]. The reaction was carried out at room temperature overnight. Epichlorohydrin was coupled to the hydroxyl end of DEG spacer, which was achieved by immersing the nonwoven into 3% epichlorohydrin–water mixture with 1 M sodium hydroxide [23] and adding drops of epichlorohydrin to the reaction periodically

at 1% per hour for 4 h. Primary amine was attached to the epoxy end of the epichlorohydrin by the immersion of the nonwoven into ammonium hydroxide (29%) at 50 °C for 8 h. Water and methanol were used to wash the sample after each of the above reactions.

2.4. Equilibrium BSA adsorption isotherms under static and flow conditions

Bovine serum albumin (BSA) equilibrium adsorption isotherms under the static condition were measured for the nonwovens modified with tertiary amine, diethylene glycol and diethylene glycol–epichlorohydrin–primary amine in PBS buffer (pH 7.4 and salt concentration: 150 mM). The amount of BSA adsorbed was determined by the differences in the concentrations of BSA solutions before and after incubation at room temperature overnight.

Equilibrium adsorption measurements of BSA binding to modified nonwoven fabric samples under the flow condition were performed by pulse experiments. Modified nonwovens were cut into discs and packed into a 1.1 cm diameter adjustable volume Vantage-L column (Millipore, Billerica, MA). The column was mounted to a Waters 616 LC system (Waters Corporation, Milford, MA) and equilibrated by passing the PBS buffer overnight. Injections of 2 ml BSA solution of 10 mg/ml concentration were made at flow rate of 0.05 ml/min or superficial velocity of 0.06 cm/min. When the binding capacity of the column was reached, the column was washed with PBS until the effluent concentration fell back to a baseline value. The effluent concentration of BSA was measured with UV absorbance at 280 nm by a Waters 2487 UV detector (Waters Corporation, Milford, MA). Bound BSA was eluted with 1 M NaCl in PBS. The dead volume of the system was determined by acetone injections.

2.5. Pulse experiments

A 1.1 cm diameter adjustable volume Vantage-L column, packed with 30 layers of PP nonwoven grafted with polyGMA and coupled with diethylene glycol, was prepared for pulse experiments. The experiments were conducted at room temperature on a Waters 616 LC system with a 200 μ l sample loop and waters 2487 UV detector. The height of the packed column was 0.60 cm. Aqueous acetone of 0.5% (v/v) and BSA solution of 1 mg/ml were injected at four different superficial velocities between 0.06 and 1.2 cm/min. The extra-column volume was accounted for by conducting the same measurements without the membrane column.

2.6. Flow permeability measurements

Pressure drops along the length of a column packed with 150 layers of modified PP nonwoven were measured at 8 different superficial velocities between 2.4 and 16.5 cm/min of PBS. The nonwoven layers were grafted with polyGMA and coupled with diethylene glycol. The corresponding column height was 3.1 cm. A peristaltic pump (Dynamax Rainin, Oakland, CA) was used to control the flow rate. The actual flow rates were determined by collecting the volume of the effluent for 1 min at the outlet of the column. Pressures at the column inlet were measured using digital pressure gauge with an accuracy of ± 0.02 bar (Ashcroft Inc., Stratford, CT). The outlet was kept at atmosphere pressure. The extra-column pressure drop was accounted for by conducting the same measurements without membrane column.

2.7. Surface analysis

The conformity of polyGMA grafts on the PP fibers was studied by Field Emission Scanning Electron Microscopy (FESEM) (JEOL 6400F, Tokyo, Japan). Before analysis, grafted nonwoven samples

were fixed on sample holder by carbon tape and sputter coated with Pd/Au alloy. The cross-sections of the fibers were measured by mounting the sample on a tilted sample holder at 45°. Benzophenone adsorption to the pretreated nonwoven fiber surface was measured by FTIR-ATR (Thermo-Nicolet Nexus 470-FTIR, Waltham, MA) with a germanium crystal. The IR penetration depth into the sample was ~ 1 μ m. The benzophenone adsorption to the surface was determined by the ratio of the height of benzophenone's aryl ketone ($\text{Ar} > \text{C}=\text{O}$) peak at 1280 cm^{-1} relative to a reference peak, the methyl ($-\text{CH}_3$) at 1380 cm^{-1} .

Grafting efficiency is conventionally measured by weight gain. However, this method is not a method suitable for these UV-grafted nonwoven fabric samples. The weight of the nonwoven increases due to grafting, but it also tends to decrease as the result of a loss of fibers during washing. The weight loss is mainly from the small diameter and short fibers that can be washed off from the nonwoven. As a result, grafting efficiency is generally underestimated when estimated by weight gain. The method to determine grafting efficiency used here is based on the ratio between the heights of two characteristic IR peaks (1380 cm^{-1} for $-\text{CH}_3$) and 1450 cm^{-1} for $-\text{CH}_2-$). This method provides a more accurate representation of the amount of grafted monomer. Characteristic $-\text{CH}_3$ to $-\text{CH}_2-$ peak ratios were measured with five samples of the original PP nonwoven and five samples of pure polyGMA. The average value of the $-\text{CH}_3$ to $-\text{CH}_2-$ peaks determined for pure polypropylene was $R_{\text{pp}} = 1.70$, and for bulk polyGMA it was $R_{\text{polyGMA}} = 0.44$. Samples of polyGMA grafted PP showed values of $-\text{CH}_3$ to $-\text{CH}_2-$ peaks between R_{pp} and R_{polyGMA} . A linear relation can be assumed for the relation between the peak ratio of a grafted sample, R_g , and the grafting efficiency α ,

$$R_g = R_{\text{pp}}(1 - \alpha) + R_{\text{polyGMA}}\alpha \quad (1)$$

Using this relation, the grafting efficiency can be estimated from the measured peak ratio R_g

$$\alpha = \frac{1.70 - R_g}{1.26} \times 100\% \quad (2)$$

The amount of tertiary amine in the nonwoven was determined by the elemental analysis (Perkin Elmer 2400 series, Norwalk, CT). The amount of primary amine was measured by a colorimetric method using picric acid [24]. Concentrations of BSA were determined by BCS assay (Pierce BCA protein Assay, Thermo-Scientific, Rockford, IL).

3. Results and discussion

3.1. Grafting mechanism

A general depiction of the UV pretreatment–UV grafting (UV–UV) method is shown in Fig. 1. The pretreatment with UV light of 185/254 nm generate carbonyl, hydroxyl and other polar groups on the fiber surface. The roughness of the surface may also increase as a result of UV exposure. These changes in surface properties tend to increase the adsorption of benzophenone from the grafting solution to the fiber surface in the UV grafting step. Consequently, both grafting conformity and grafting rates increase with pretreatment time.

In Fig. 2, the XPS results show the change of oxygen (O1s) content in the nonwoven as a function of pretreatment time. The O1s peak is associated with the formation of polar groups on the surface of the PP nonwoven. The observed split of the O1s peak is due to the presence of differential charges on the surface. The results show that the total area of O1s peaks around a binding energy of 530 eV increase with the pretreatment time. This provides strong evidence that the concentrations of polar groups on the surface increase with the pretreatment time. The mechanism for

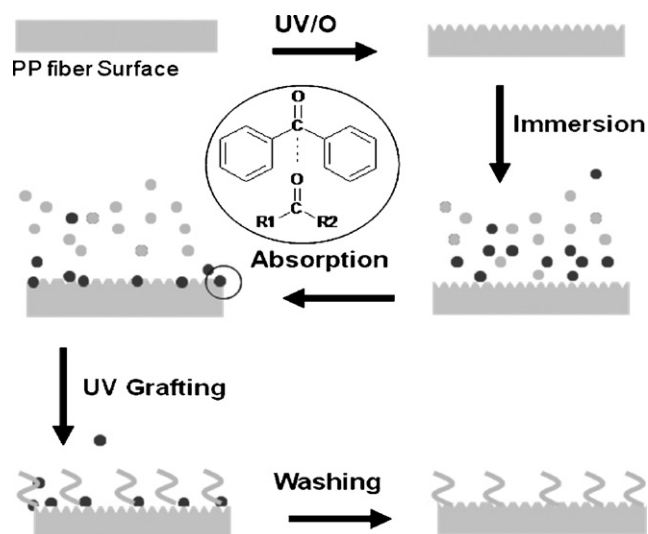


Fig. 1. Schematic illustration of the UV-UV procedure for the surface grafting of polyGMA on PP nonwoven fibers. Dark dot: benzophenone; grey dot: GMA monomer.

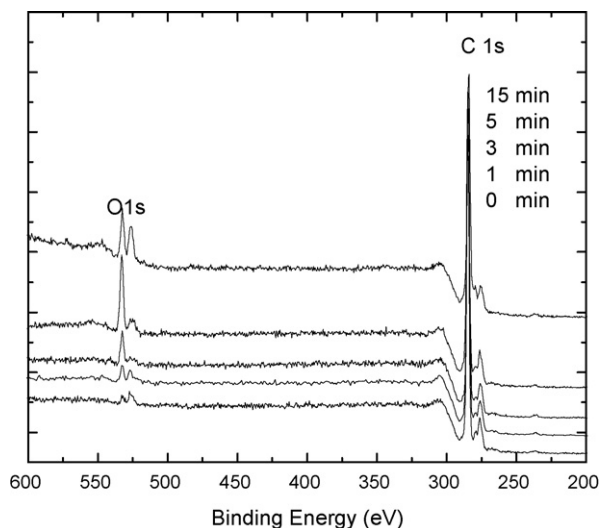


Fig. 2. XPS of PP nonwovens after the UV pretreatments for different times. The curves from the top to the bottom of the figure correspond to 15–0 min of pretreatment.

the generation of polar compounds by UV oxidation is well understood [25]. Commercial PP and other polyolefins contain small amounts of unsaturated groups, carbonyls, hydroperoxides and catalyst residues. These compounds act as initiators that are excited

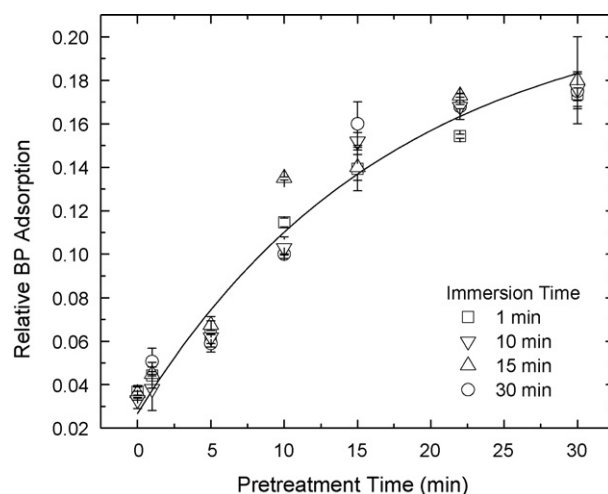


Fig. 4. Relative BP adsorption as a function of UV pretreatment time measured at different immersion times.

and cleaved under UV exposure. During the process, more peroxide and carbonyl compounds are generated from oxidation of PP by ozone, which is produced from the interaction between oxygen and the UV, and lead to more cleavages. Direct results of this process include an increase in the concentration of carbonyl-containing compounds such as esters, acid, ketone, aldehyde, lactone as well as hydroxyls [26]. We suspect that the surface roughness may also have increased with pretreatment time as a result of the cleavage of the polymer chains and formation of new surface compounds. However, the SEM used for this study cannot clearly resolve the roughness change. It is suspected that roughness changes on the fiber surface will have a scale of a few nanometers, while the SEM can only resolve structures of around 20 nm.

The surface pretreatment with UV can significantly improve the conformity of polyGMA grafts. In Fig. 3 two nonwovens, both heavily grafted under the same grafting conditions, exhibit very different conformities depending on the pretreatment. The non-woven without the pretreatment shows non-conformal grafts that protrude into nearby voids and block pores, while the pretreated sample shows uniform and conformal grafts on the fiber surfaces.

The improvement in conformity is attributed to the enhanced adsorption of benzophenone as a result of the pretreatment. Fig. 4 shows the results of benzophenone (BP) adsorption measured by immersing the pretreated nonwovens into a 1.3% benzophenone in butanol solution without simultaneous UV irradiation. The relative BP adsorption values, which are the ratios of the characteristic IR peak (1280 cm^{-1}) to the reference peak (1380 cm^{-1}), are plotted as a function of pretreatment time. The standard error was estimated from data measured at different spots on the same spec-

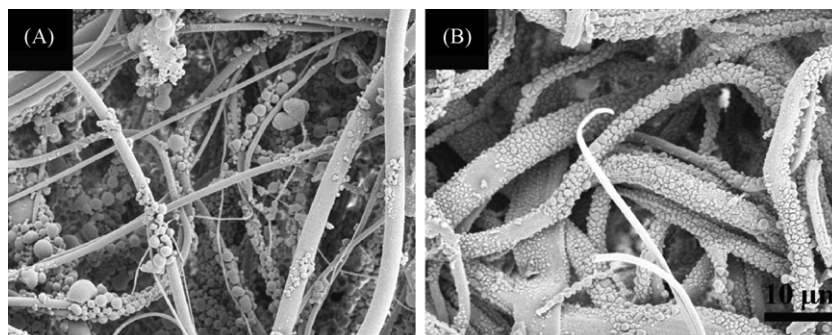


Fig. 3. Grafting with 20% GMA solution for 30 min (A) without pretreatment; (B) with pretreatment of 15 min. The scale for images (A) and (B) is shown in (B).

imen. It is evident that as the pretreatment time is increased from 0 to 30 min, the adsorption of BP increases correspondingly. The curves indicate a strong dependence of BP adsorption on the pretreatment time. This is attributed to the effects of increase in the concentration of polar groups and probable increase in the roughness. Dipole–dipole interactions between benzophenone and newly formed polar groups resulting from pretreatment allow the adsorption of benzophenone to the fiber surfaces. Without UV pretreatment, the adsorption of benzophenone to the fibers was minimal. Any possible increases in surface roughness of fibers could also provide additional surface area for benzophenone adsorption. More importantly, despite the use of various immersion times,

adsorption curves overlap into a single curve within the experimental error. This result implies that the adsorption of benzophenone was very fast, and equilibrium of BP was quickly established between the solution and the pretreated fiber surface upon contacting. In separate experiments, it was shown that BP can be removed from the nonwoven by simply dipping into pure butanol, indicating that BP is not covalently coupled to the fiber surfaces as a result of the pretreatment.

Even though BP adsorption shows a strong dependence on the pretreatment time, the amount of adsorption does not increase linearly with time. Instead, it increases rapidly at short pretreatment times, but grows slowly after long pretreatment times. Eventually,

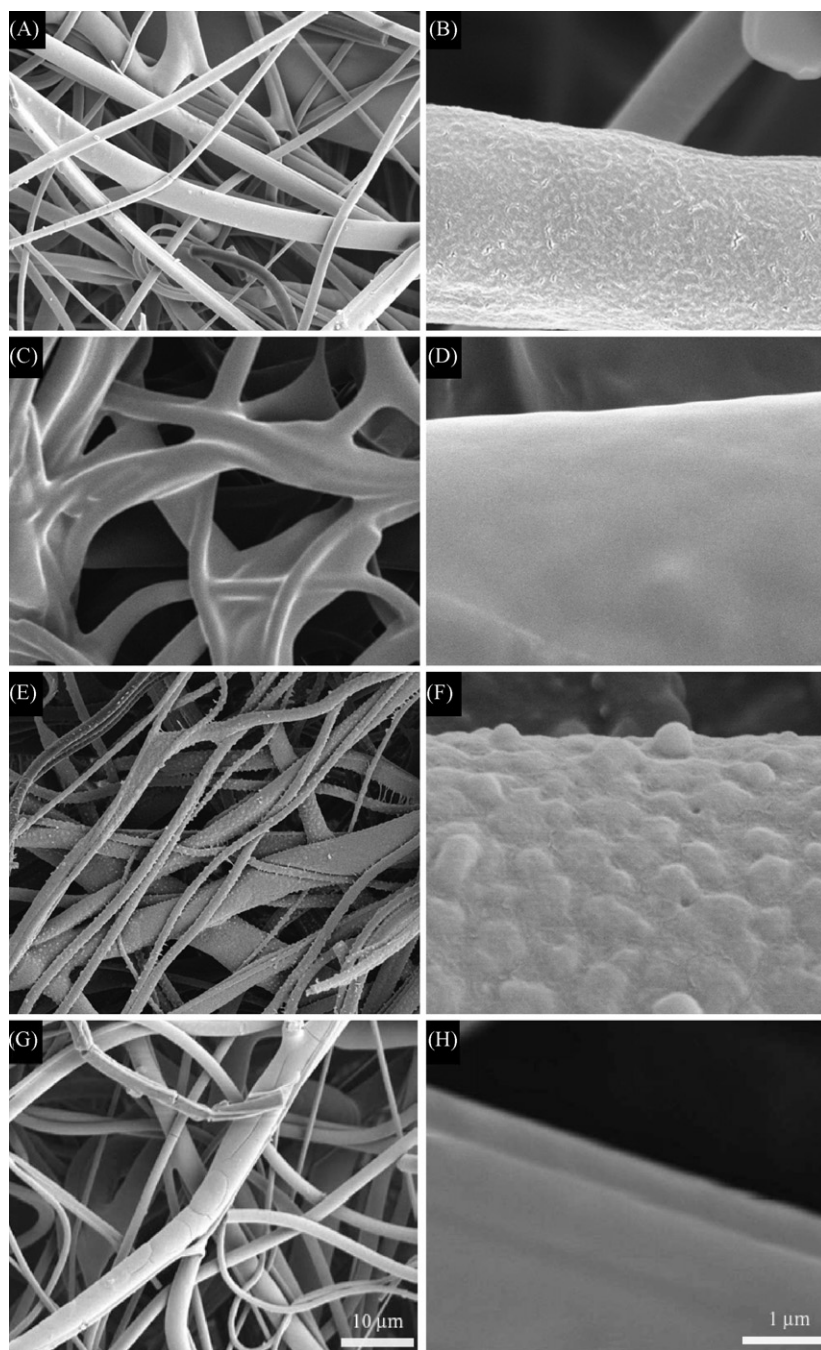


Fig. 5. (A) Polypropylene nonwoven before grafting; (B) fiber surface before grafting; (C) grafted polypropylene nonwoven before washing; (D) grafted fiber surface before washing; (E) grafted polypropylene nonwoven after washing; (F) grafted fiber surface after washing; (G) grafted polypropylene nonwoven with high initiator to monomer ratio after washing; (H) grafted fiber surface with high initiator to monomer ratio after washing. The scale for images (A), (C), (E), and (G) is shown in (G); that for images (B), (D), (F) and (H) is shown in (H).

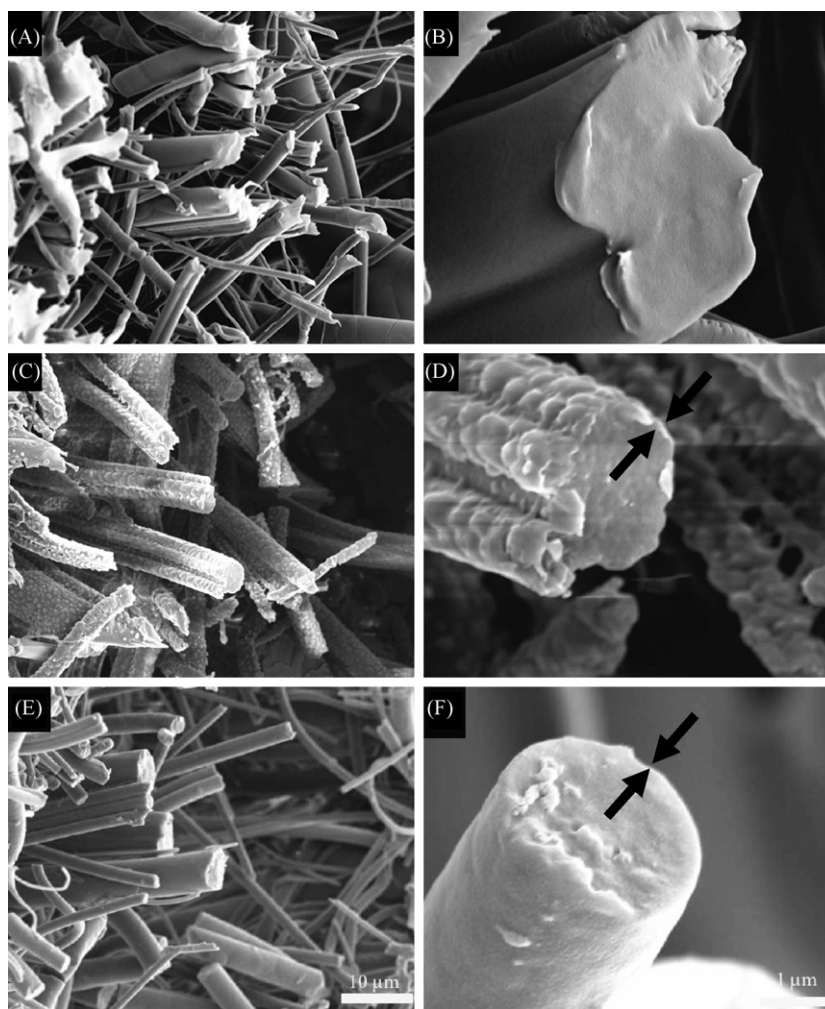


Fig. 6. (A) Cross-section of original PP nonwoven; (B) cross-section of a single original PP nonwoven fiber; (C) cross-section of nonwoven after grafting at I:M = 1:25; (D) cross-section of a single fiber grafted at I:M = 1:25; (E) cross-section of nonwoven after grafting at I:M = 1:5; (F) cross-section of a single fiber grafted at I:M = 1:5. The scale for images (A), (C) and (E) is shown in (E); that for images (B), (D) and (F) is shown in (F). Arrows indicate the graft layer.

the adsorption of BP approaches a limited value that depends on the effects of polar groups and possibly increased roughness due to the pretreatment.

3.2. Conformity and uniformity of grafts

Very conformal polyGMA grafts can be attained on PP nonwoven fibers by the UV–UV method. The porous structure of the PP nonwoven and the surface of nonwoven fibers before and after grafting are shown in Fig. 5. Fig. 5A and B illustrates the porous structure and fiber surface of the nonwoven before grafting. In Fig. 5A, it can be seen that randomly oriented nonwoven fibers form interconnected pores. Fig. 5B shows the fiber surface scattered with cracks resulting from the melt-blowing process. Fig. 5C and D shows the same nonwoven fabrics after grafting, but before washing. The grafting conditions for samples shown in this figure included 15 min of pretreatment and 15 min for surface grafting at initiator to monomer ratio 1:25. It can be seen that the fiber surface is covered by a smooth film, consisting of polyGMA grafts, polyGMA homopolymers, benzophenone and unreacted GMA monomers. Ungrafted materials tend to form menisci between fibers. As a result, the pore size is reduced. Washing by sonication in THF and methanol removed ungrafted molecules from the fiber surface and pores. In Fig. 5E and F, the nonwoven recovers its microstructure after washing. The only difference between the grafted and original is that the

fiber surface is covered with conformal, undulating polyGMA grafts. When the initiator to monomer ratio in the grafting was increased from 1:25 to 1:5, the morphology of the grafts changed from very rough to very smooth as shown in Fig. 5G and H. It is suspected that this phenomenon is due to the decrease of graft molecular weight and weight distribution. As the initiator concentration is increased, both initiation and termination rates are increased, and the polymer molecular weight is decreased [27]. As a result, the number of large particles of polyGMA with large molecular weights is reduced.

The cross-sections of nonwoven fibers before and after grafting are shown in Fig. 6. Fig. 6A and B shows the cross-sections of the original PP nonwoven in Fig. 5A and B. It can be seen that some fibers have complex shaped cross-sections, which deviate from round circles. Fig. 6C and D shows the cross-sections of the grafted nonwoven in Fig. 5E and F. The images clearly demonstrate that coarse and very conformal grafts of polyGMA has been obtained on the fibers of either round or complex shaped cross-sections. Fig. 6E and F shows the cross-sections of the grafted nonwoven in Fig. 5G and H. In this case, smooth and very conformal grafts are exhibited. Even though the thickness of the graft layer is uniform, an accurate quantitative thickness measurement is difficult to obtain because the low contrast between the grafted and the original part of the fiber in the images. It is estimated in the range of 100–200 nm.

Even if the UV light was applied on the nonwoven only from one direction, its effect was not limited to the fiber surface that

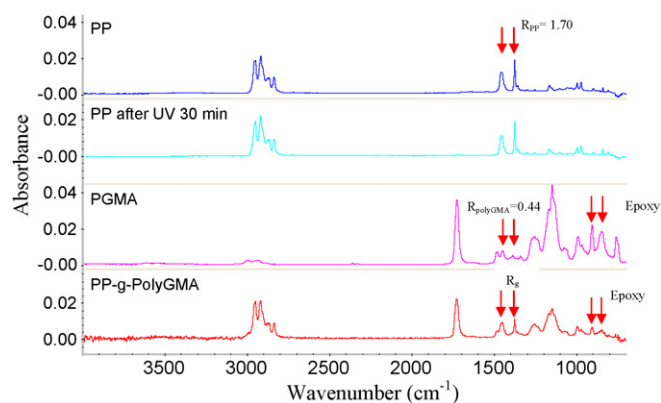


Fig. 7. IR spectra of original PP nonwoven, PP nonwoven after pretreatment, pure polyGMA and polyGMA grafted PP. Arrows indicate the characteristic peaks for epoxy and for the calculation of grafting efficiency.

is directly exposed to the UV. As can be seen in the images, surface exposed to various directions were modified as well. This is due to the fact that in a porous system a large portion of incident light is reflected and scattered at all directions from all surfaces [28]. Therefore, the surface not directly exposed to the direction of the incident light can be illuminated. However, the intensity of the UV light decays inside the nonwoven, which can lead to spatial variation of grafting for thick samples. Further work has shown that gradients in grafting levels can be mitigated by treating the samples by UV from multiple directions.

The IR spectra of pure PP nonwoven, PP nonwoven after pretreatment, polyGMA and polyGMA grafted PP nonwoven are shown in Fig. 7. The appearances of the carbonyl peak ($>C=O$) at 1720 cm^{-1} and epoxy peaks at 845 and 905 cm^{-1} in the grafted PP sample are clear indications of polyGMA grafts. The UV pretreated sample also shows a small peak at 1720 cm^{-1} , associated with carbonyl compounds formed by the UV pretreatment. Even though no clear hydroxyl peak (3300 cm^{-1}) is shown for the pretreated sample, water can wet the surface better after pretreatment, an indication of an increase in surface hydrophilization, possibly due to the formation of surface hydroxyl groups. Characteristic peaks at 1380 cm^{-1} and 1450 cm^{-1} are also indicated for PP, polyGMA and polyGMA grafted PP in the figures. The ratio of these two peaks of the grafted PP nonwoven was used to calculate the grafting efficiency.

3.3. Grafting kinetics

The enhanced adsorption of BP improves not only the conformity of grafts, but also the grafting rate. Grafting kinetic curves at different pretreatment times for grafting with 20% GMA and 1.3% benzophenone are shown in Fig. 8. Curves of high grafting efficiencies are associated with high grafting rates. Clearly, the grafting rate increases with pretreatment time for short times, then levels off to a constant rate. As shown in Fig. 8, the grafting curve for 10 min of pretreatment overlaps that for 15 min of pretreatment. However, the adsorption of BP for 10 min of pretreatment is lower than that for 15 min of pretreatment (Fig. 4). It can be concluded that the grafting rate does not always depend on the amount of initiator on the surface. This phenomenon has also been observed in polymer grafting on flat surfaces [29,30]. It has been hypothesized that not all the initiators on the surface contribute to graft insertion because of steric effect inhibition from nearby grafts. There exists a cutoff initiator density beyond which the grafting rate does not increase significantly. Here, the cutoff density lies at a relative BP adsorption of ~ 0.10 . It is also interesting to note that in Fig. 8 all the curves show a lag period of ~ 5 min. This lag period is caused by the pres-

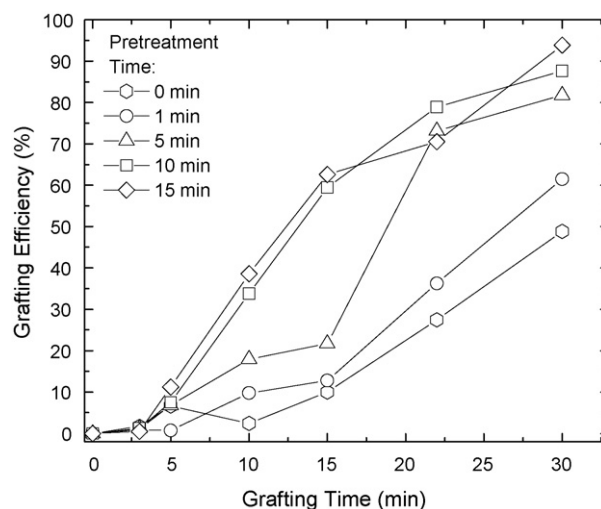


Fig. 8. Grafting efficiency as a function of grafting time measured at different pretreatment times.

ence of oxygen trapped in the solvents used in these experiments, which were not degassed.

3.4. BSA adsorption isotherms for the nonwovens coupled with tertiary amine, diethylene glycol and diethylene glycol-epichlorohydrin-primary amine

The structures of the surfaces after the coupling of tertiary amine, diethylene glycol and diethylene glycol-epichlorohydrin-primary amine are illustrated in Fig. 9. The density of tertiary amine was measured as $1.3 \times 10^{-3}\text{ mol/g}$ by elemental analysis and that of primary amine was $8.0 \times 10^{-6}\text{ mol/g}$ obtained from a colorimetric method using picric acid. The orders of magnitude difference in amine density between the tertiary amine and the primary amine comes from the fact that the reaction between diethylene glycol and epichlorohydrin has a very low conversion. Upon conversion to primary amines by treatment with ammonium hydroxide, this leads to a much lower density of amines. The tertiary amine was attached directly to polyGMA grafts without spacer, resulting in a much higher amine density. At neutral pH, both tertiary and primary amines are positively charged, which can absorb negatively charged proteins, such as BSA, through electrostatic interactions. On the other hand, diethylene glycol spacer should incur little protein binding due to its steric and hydrophilic effects.

BSA was selected as a model protein to illustrate the differences in the adsorption properties of the three surfaces, which are shown in Fig. 10. Adsorbed BSA is plotted as a function of equilibrium BSA concentration for static binding measurements. As shown in the figure, the surface with tertiary amines exhibits a significant BSA adsorption of 122 mg/g at 6.4 mg/ml BSA concentration. The isotherm also shows a plateau at approximately 50 mg/g of adsorbed protein concentration. This plateau does not correspond to a region of monolayer adsorption on the fibers. Monolayer saturation of BSA on an interface has been reported between 2 mg/m^2 and 9 mg/m^2 , depending on the orientation of the protein on the surface [31,32]. The PP nonwoven used in this study has surface area of $0.3\text{--}0.5\text{ m}^2/\text{g}$. Therefore, if BSA were adsorbed to the PP nonwoven in a monolayer, the value for the isotherm should be less than 5 mg/g . The adsorption plateau for the surface modified with tertiary amine shows a value significantly higher than the monolayer saturation of the PP nonwoven. This plateau is probably associated with a saturated multilayer adsorption. This result is similar to the result from a previous study by other authors [32] in which

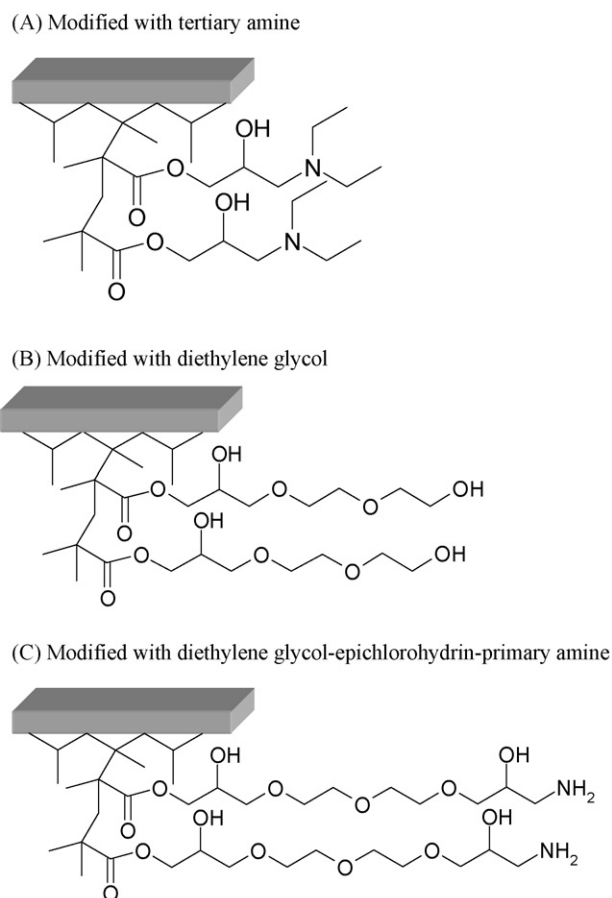


Fig. 9. Molecular structures of polypropylene surface modified with (A) polyGMA and coupled with tertiary amine, (B) polyGMA and coupled with diethylene glycol, and (C) polyGMA and coupled with diethylene glycol, epichlorohydrin and primary amine.

BSA adsorption as high as 490 mg/g was reported and explained as multilayer packing of BSA on a tertiary amine modified polyethylene hollow fiber membrane. The membrane used for that study had approximately 10 times more surface area per gram of material than the PP nonwoven used in this study. With increasing BSA concentration, the protein adsorption to the nonwoven shows a

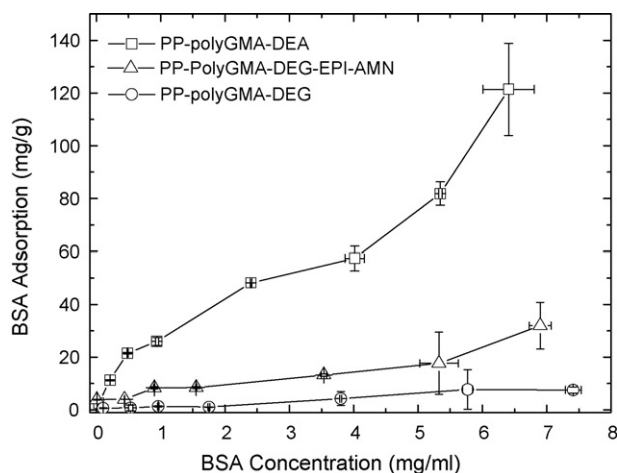


Fig. 10. BSA adsorption isotherms for PP nonwoven grafted with polyGMA and coupled with tertiary amine, PP nonwoven grafted with polyGMA and coupled with diethylene glycol and PP nonwoven grafted with polyGMA and coupled with diethylene glycol, epichlorohydrin and primary amine.

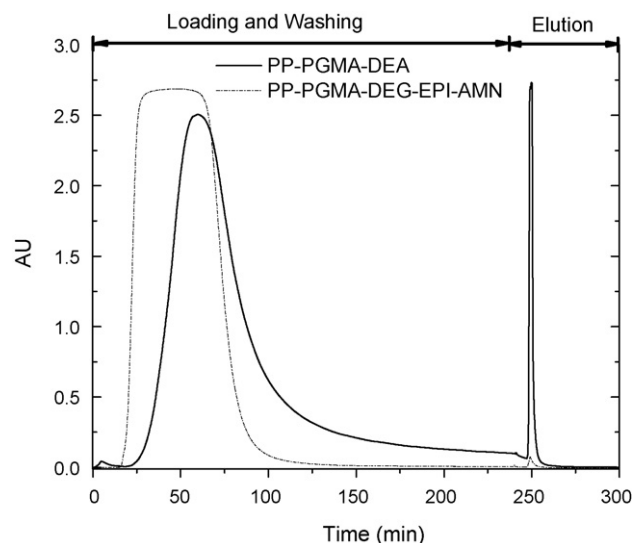


Fig. 11. Binding of BSA on columns packed with nonwovens modified with tertiary amine and diethylene glycol-epichlorohydrin-primary amine.

rapid monotonic increase. Similar results have been reported for BSA and other proteins adsorbed at the gas-liquid interfaces [31]. This rapid increase in adsorption can be attributed to fast multilayer formation at high protein concentrations.

Nonwoven fabric modified with diethylene glycol shows little binding of BSA. This is the result of hydrophilization and steric effects of diethylene glycol. Poly(ethyleneglycol) has been widely used for modification of polymer surfaces to reduce nonspecific protein binding [33–35]. There is a slight increase in adsorption when BSA concentrations are above 4 mg/g.

The nonwoven modified with diethylene glycol spacer, epichlorohydrin and primary amine shows enhanced binding to BSA. This result indicates that primary amines can function as weak anion exchangers. The isotherm curve in this case shows a plateau at 7 mg/g when the BSA concentration in solution is less than 2 mg/ml. The plateau value is close to the value for monolayer saturation. At higher concentrations, BSA adsorption on the nonwoven modified with primary amine also exhibits multilayer adsorption behavior.

Chromatograms for equilibrium binding of BSA by the nonwoven modified with tertiary amine and primary amine under flow conditions are shown in Fig. 11. Both materials show that the BSA can be eluted by 1 M PBS buffer. However, the binding capacity for the nonwoven modified with tertiary amine is much higher than that for the nonwoven modified with primary amine.

The binding capacity for the tertiary amine nonwoven membrane is 102 mg/g with the BSA loading concentration of 10 mg/ml, which is about 85% of its static binding capacity for the same loading concentration. The binding capacity for primary amine is 3 mg/g at the loading concentration of 10 mg/ml, which is only 8% of its static binding capacity. Since the nonwoven fibers are not porous, diffusional limitations should not be the major cause for the difference between the static and flow condition binding capacities. We attribute the difference between the binding capacities under static and flow conditions to a fraction of adsorbed BSA that is only loosely attached to the nonwoven and can be washed away in the binding buffer. Because in the static condition tests the washing step was not applied, the measured amount of adsorbed BSA included those bound by ion exchange as well as by other, weaker forces. Under the flow condition, the nonwoven experienced a continuous flow of the protein solution as well as a washing step. As a result, the accumulation of loosely attached BSA on the nonwoven was sig-

nificantly reduced. Therefore, the measured equilibrium binding capacity under the flow condition is lower than that obtained under the static condition. Using a loading BSA concentration of 10 mg/ml, the loosely bound BSA contributes approximately 20 mg/g to the equilibrium binding capacity of the nonwoven modified with tertiary amine and approximately 32 mg/g to that of the nonwoven modified with primary amine, measured under the static condition.

3.5. Porosity measurements

The interstitial and total porosities of a packed column can be measured by the first absolute moments of pulse injections of different size tracers under nonbinding conditions [36]. Acetone is commonly used as tracer to measure the total porosity; and a macromolecule of size larger than the intra-particle pore sizes can be used to determine the interstitial porosity for that particular macromolecule. The first absolute moment is defined by the expression

$$\mu_1 = \frac{\int_0^\infty C(L, t) t dt}{\int_0^\infty C(L, t) dt} \quad (3)$$

In the equation, μ_1 is the first absolute moment, C is the concentration of tracer exhibiting the column and t is time. The relation between the first absolute moment and the total porosity ε is given by

$$\mu_1 = \frac{L}{u_0} \varepsilon \quad (4)$$

In this equation, u_0 is the superficial velocity and L is the height of the column. If μ_1 is plotted versus L/u_0 , the slope of the curve is porosity.

In this study, this technique was applied to measure porosities of a nonwoven packed column. A major difference between a conventional packed-bed column and nonwoven packed column lies in the fact that the former is packed with porous particles and the latter is packed with nonporous fibers. Even though, there is no intra-fiber porosity in the same sense as intra-particle porosity, two tracers of different sizes, acetone and BSA, were used to explore the porosity difference of the nonwoven packed column, which was packed with the nonwoven consisting of fibers grafted with polyGMA and diethylene glycol. The isotherm in Fig. 10 showed that adsorption of BSA on this type of surface is negligible. Therefore, it satisfied the nonbinding conditions.

Fig. 12 shows the plot of first absolute moments versus L/u_0 for the two tracers. The results indicate that acetone has longer residence time and larger slope than BSA. This is apparently due to the fact that acetone is much smaller than BSA. The pores accessible to acetone are inaccessible to BSA. The porosity measured with acetone was 0.80, which is identical to the total porosity calculated by the density difference between solid pure polypropylene and the polypropylene nonwoven. The porosity measured with BSA was 0.48. Because the fibers are nonporous, the difference in the porosities measured by acetone and BSA could only be attributed to reasons other than voids inside fibers. The first reason may be associated with the presence of voids between the overlapping sections of fibers. A melt-blowing process was used to produce the PP nonwoven fabrics used in these studies. During this process, molten polymer fibers can fuse with each other and presumably create small gaps between fibers that are big enough to accommodate acetone, but not a molecule like BSA. It is also possible that voids may exist inside the polyGMA graft layer after the evaporation of the solvent. The voids, which are accessible to acetone but too small for BSA, account for 30% of the total volume of the nonwoven column.

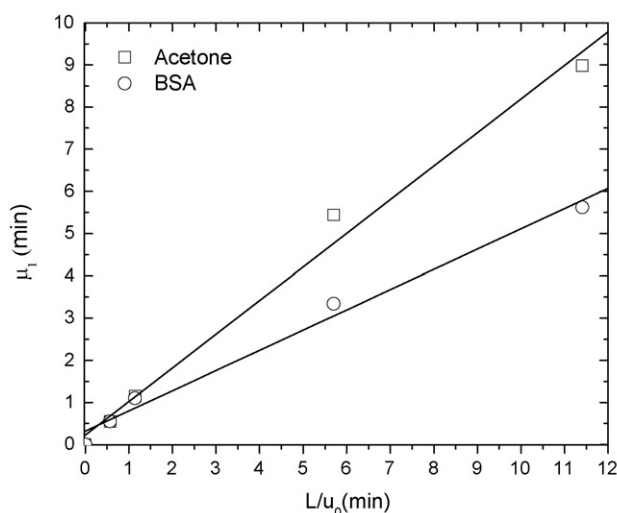


Fig. 12. First moment versus L/u_0 for pulse injects of acetone and BSA to a column packed with 30 layers of PP nonwoven modified with diethylene glycol.

3.6. Pressure drop measurements

Pre Darcy's Law can be used to describe the pressure drop in a porous medium when the Reynolds number is low [37]

$$u_0 = \frac{k}{\mu} \frac{\Delta P}{L} \quad (5)$$

In this equation, u_0 is the superficial velocity; k is the permeability coefficient; μ is the viscosity of the mobile phase, which is 0.01 poise for the PBS buffer; ΔP is pressure drop across the medium and L is the height of medium. In Fig. 13, the superficial velocity is plotted as a function of the measured $\Delta P/L\mu$ value for a column packed with 150 layers of nonwoven. The slopes of the curves are the permeability coefficients, which equal to $1.08 \times 10^{-7} \text{ cm}^2$. This value is between 1 and 3 orders of magnitude higher than the literature values of membranes developed for the similar purposes [38–40] and between 1 and 2 orders of magnitudes higher than the values of ion exchange columns packed with resins [41,42]. In this experiment, we also identified an issue associated with using adjustable volume Vantage-L column to test a stack of nonwoven membranes. The pressure drop can suddenly increase rapidly if the column is only slightly over tightened. This is probably caused by the blocking of pores ($\sim 15 \mu\text{m}$) of the bed support by fibers from

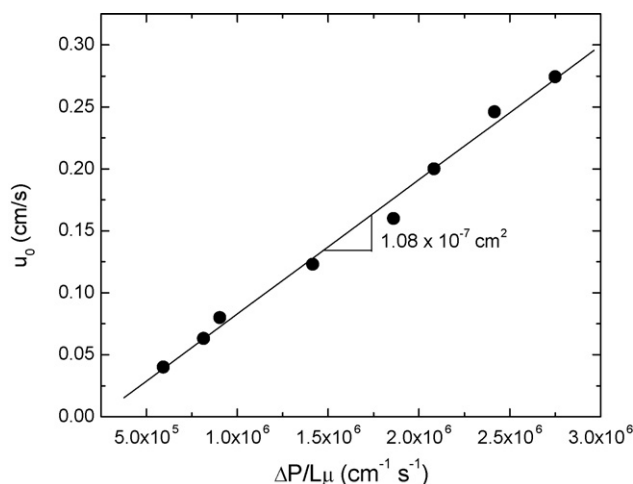


Fig. 13. Superficial flow velocity versus pressure drop for a column packed with 150 layers of PP nonwoven modified with diethylene glycol.

the nonwoven layer. If this happens, the pressure drop along the column can be overestimated.

4. Conclusions

In summary, this is the first study that describes an effective UV pretreatment and UV grafting process (UV–UV) that generates very conformal polyGMA layers of sub-micron thickness on PP nonwoven fabrics for applications in bioseparations. UV pretreatment can significantly improve the adsorption of benzophenone to the fiber surface. Enhanced benzophenone adsorption increases the conformity and grafting rate of the polyGMA coating. This activation process described in this work results in highly conformal and uniform surface coatings on PP nonwoven fabrics which can then be used to prepare useful membranes for bioseparations by ion exchange or affinity capture. In this work, reaction times for both UV pretreatment and UV grafting are relatively long. We expect that these processing times can be significantly reduced if lamps with high intensities are used, even though the processing time might not be completely inversely related to lamp intensity.

The activated nonwoven membranes with conformal, uniform polyGMA surface layers were easily modified with tertiary amines, diethylene glycol spacer, or a spacer terminated with a primary amine. These modified membranes exhibited very high protein binding capacities for BSA under static and flow conditions. The results of the pulse response experiments and pressure drop measurements indicate that the modified membranes maintain very high porosity for both small and large solutes and very high flow permeability compared to other membrane types and to packed beds of chromatographic resins. These results indicate that nonwoven fabrics membranes may become attractive alternatives to other types of membranes and particles currently being used in industrial bioseparations.

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